

Australian monetary–fiscal coordination model (Julia)

This folder contains a Julia extension of the monetary–fiscal policy game in `SI_game`. It combines a small Australian three-equation macro model with a dynamic two-player policy game and a separate Leeper-style fiscal-dominance experiment.

The distinction between **preferences** and **policy regimes** is deliberate:

- the central bank and fiscal authority each have one fixed loss function;
- those primitive weights are unchanged across Nash, cooperation, central-bank leadership and fiscal leadership;
- monetary and fiscal dominance differ only through policy-rule coefficients, especially the interest-rate response to inflation and the fiscal response to debt.

The full economic explanation is in [THREEEQUATIONMODEL.md](#). More concise implementation notes are in [MODEL_NOTES.pdf](#).

Running the model

From this directory:

```
julia --project=. -e "using Pkg; Pkg.instantiate()"
julia --project=. run_model.jl
julia --project=. test/runtests.jl
```

The first command installs the exact dependencies in `Manifest.toml`. The second uses the checked-in RBA data snapshot, estimates the Australian macro block, solves all policy games and regenerates the outputs. To download the latest RBA tables first, use:

```
julia --project=. run_model.jl --refresh-data
```

To regenerate the PDF documentation:

```
julia --project=. build_pdfs.jl
```

Julia code map

- `policy_game.jl` contains the commented state-space model, fixed preference primitives, quadratic game solver and active/passive policy-rule simulator.

- `data_pipeline.jl` downloads and parses the RBA tables, constructs gaps, implements the HP filter and estimates the reduced-form equations.
- `run_model.jl` defines the single calibration, shock experiments, explicit policy regimes, output tables and plots.
- `test/runtests.jl` tests the transitions, Nash first-order conditions, cooperative loss comparison, fixed-preference regime experiment and data fit.
- `src/AustralianMonetaryFiscalGame.jl` is the small package entry point needed for reproducible Julia project loading and precompilation.
- `build_pdfs.jl` converts the three Markdown notes to PDF locally.

Data

All series are public and are downloaded from the [RBA Statistical Tables](#).

Table	Series ID	Use
H1	GGDPCVGDGP	Chain-volume real GDP and HP-filtered output gap
H2	GGDPECCVPD	Chain-volume public demand and fiscal-policy proxy
G1	GCPIOCPMTMQP , GCPIOCPMTMYP	Trimmed-mean inflation and ex-post real rate
F1.1	FIRMMCRT	Monthly cash-rate target, averaged to quarters

The estimation window is 1993 Q1–2019 Q4. Ending before the pandemic avoids asking one constant-coefficient model to explain both the effective-lower-bound and lockdown periods. The checked-in raw files, source URLs, byte counts and SHA-256 hashes are recorded in `data/raw/manifest.toml`.

This is a **data-informed calibration**, not causal identification. The cash rate and public demand are endogenous policy choices, and the HP-filtered gaps are two-sided estimates.

Fixed objective primitives

Authority	Inflation	Output	Debt	Instrument adjustment	Discount
Central bank	1.00	0.25	0.00	0.12	0.99
Fiscal authority	0.25	1.00	0.45	0.08	0.99

These illustrative weights are defined once in `run_model.jl` and written to `output/model_primitives.toml`. The cooperative planner's 50–50 aggregation of the two fixed losses is part of that solution concept; it does not change either authority's primitive loss function.

Generated outputs

- `output/australian_inputs.png` : Australian data used by the model.
- `output/demand_shock_game.png` : four solution concepts after a demand shock.
- `output/supply_shock_game.png` : four solution concepts after an inflation shock.
- `output/fiscal_dominance_comparison.png` : active/passive policy-rule regimes, evaluated with the same objective functions.
- `output/game_summary.csv` : loss and peak-response comparison for the games.
- `output/rule_regime_summary.csv` : rule coefficients, losses and terminal debt.
- `output/model_primitives.toml` : the complete fixed calibration.
- `output/estimated_coefficients.csv` and `estimation_summary.toml` : model fit.
- `output/policy_sensitivities.csv` : first Nash action's response to each state.
- `data/processed/australia_quarterly.csv` : analysis-ready quarterly data.

Main limitations

The strategic game is a finite-horizon, perfect-foresight open-loop game. The fiscal-dominance experiment is a backward-looking active/passive rule simulation. It illustrates the mechanism but is not a rational-expectations FTPL determinacy proof. A literal FTPL extension requires a forward-looking IS curve, New Keynesian Phillips curve, nominal government valuation equation, surplus rule and equilibrium-selection conditions.